

VisionARies: Bringing Tradition and Technology through Augmented Reality (AR) Heritage Games

Nur Sahira Batrisyia
Faculty of Computer Science
& Information Technology
Universiti Malaya
Kuala Lumpur, Malaysia
23001656@siswa.um.edu.my

Muhammad Aqlan
Faculty of Computer Science
& Information Technology
Universiti Malaya
Kuala Lumpur, Malaysia
23002551@siswa.um.edu.my

Ameer Zafran
Faculty of Computer Science
& Information Technology
Universiti Malaya
Kuala Lumpur, Malaysia
23002223@siswa.um.edu.my

Muhammad Hariz
Faculty of Computer Science
& Information Technology
Universiti Malaya
Kuala Lumpur, Malaysia
23002735@siswa.um.edu.my

Anis Atirah
Faculty of Computer Science
& Information Technology
Universiti Malaya
Kuala Lumpur, Malaysia
23002257@siswa.um.edu.my

Edlan Nabil
Faculty of Computer Science
& Information Technology
Universiti Malaya
Kuala Lumpur, Malaysia
23002373@siswa.um.edu.my

Garry Fresco
Faculty of Computer Science
& Information Technology
Universiti Malaya
Kuala Lumpur, Malaysia
23001954@siswa.um.edu.my

Najah Safwah
Faculty of Computer Science
& Information Technology
Universiti Malaya
Kuala Lumpur, Malaysia
23001748@siswa.um.edu.my

Nurin Allya
Faculty of Computer Science
& Information Technology
Universiti Malaya
Kuala Lumpur, Malaysia
23002635@siswa.um.edu.my

Kearney Leith Julius
Faculty of Computer Science
& Information Technology
Universiti Malaya
Kuala Lumpur, Malaysia
23001894@siswa.um.edu.my

Abstract— *The survival of local heritage games in Malaysia is currently threatened by low public engagement and limited volunteer support. This SULAM-based project, titled "VisionARies," proposes using Augmented Reality (AR) and social media platforms like TikTok to increase visitor engagement and cultural learning at Badan Warisan Malaysia (BWM). By collaborating with Badan Warisan Malaysia (BWM), the team aims to provide an interactive, AR-guided experience that educates tourists on the rules and history of traditional games. This approach aims to reduce the operational workload of site staff while ensuring consistent and engaging knowledge delivery. The project follows a 14-week development timeline, moving from initial planning, design, development, testing and deployment phase. By using AR and social media*

as a medium, VisionARies seeks to achieve a greater appreciation of cultural heritage among younger generations while preserving sustainability of heritage sites.

Keywords— *Augmented Reality, Cultural Heritage, Traditional Games, Visitor Engagement, Heritage Education*

SULAM types— *Hand Holding*

I. INTRODUCTION

Malaysian traditional heritage games are facing declining relevance due to low public awareness, limited volunteer support and lack of people-powered heritage site operations. At Badan Warisan Malaysia (BWM), visitor engagement relies heavily on conventional guided explanations and static

exhibits, which restrict scalability and consistency. In response to this challenge, VisionARies project was developed as a SULAM service-learning initiative that integrates Augmented Reality (AR) and social media to create an interactive educational platform for heritage appreciation. The project involves undergraduate students from Universiti Malaya enrolled in the WIX2001 course. Students assume defined roles such as Project Manager, AR Developer, UI/UX Designer, Research and Content Lead, and Testing Coordinator. The primary community collaborator is Badan Warisan Malaysia (BWM), a national heritage organization responsible for preserving and promoting Malaysian cultural heritage. The beneficiaries of the project include local visitors, tourists, and online users who can access the AR experience both on-site through QR codes and remotely via TikTok. VisionARies is driven by three primary objectives which are to reduce the operational workload of heritage site staff through automated AR-based guidance, to increase visitor awareness and understanding of Malaysian traditional heritage games and to significantly enhance visitor engagement through interactive and experiential learning using AR technology. This project operates within the domain of augmented reality, cultural heritage preservation, digital learning and service-learning (SULAM), focusing on the development of an educational prototype rather than a commercial product. It also aligns with several United Nations Sustainable Development Goals (SDGs) particularly SDG 4 on Quality Education through innovative learning methods, SDG 11 on safeguarding intangible cultural heritage, SDG 9 on technological innovation, and SDG 17 through university–community partnership. This project is structured into a systematic 14-week timeline divided into four phases which are project conception and initiation (Weeks 1–3), project definition and planning (Weeks 4–6), project development and testing (Weeks 7–11) and project launch and presentation (Weeks 12–14). Through this approach, VisionARies aims to demonstrate how digital platforms can support sustainable heritage education while fostering a strong collaboration between students and the community.

II. LITERATURE REVIEW

The preservation of intangible cultural heritage like traditional games increasingly depends on digital tools that engage users actively. Literature establishes Augmented Reality (AR) as an effective

medium for immersive learning at heritage sites, while social platforms like TikTok are recognized for enabling participatory cultural communication. However, a gap exists in designing social AR filters that deliberately develop users' thinking and communication skills through interactive gameplay. This review outlines these foundations and positions the VisionARies project which is a TikTok AR filter for Malaysian traditional games as an initiative that bridges this gap by integrating skill-based interaction with cultural engagement.

A. Domain Overview: AR Filters as Tools for Interactive Heritage Engagement

Recent literature extends the use of Augmented Reality (AR) beyond museum-based apps to social media AR filters and lenses, recognizing their potential for mass, playful engagement with cultural content. These filters lower the barrier to entry, allowing users to interact with heritage elements in personal, creative spaces (Ruiz, 2021). This project's domain sits at the intersection of gamified AR, intangible cultural heritage (ICH) preservation, and digital communication skills.

Applied Thinking Skills via Gamified Interaction: Research in interactive media shows that even simple, gesture-based interactions like tapping can foster procedural understanding and cause-effect reasoning (Ibrahim, 2018). In a heritage context, an AR filter that requires users to tap to bounce a *sepak takraw*, spin a *gasing*, or lift a *wau* translates abstract cultural artifacts into interactive systems. Users must observe, time actions, and learn the virtual object's physics, applying basic problem-solving and hand-eye coordination to “master” the traditional game's core kinetic action.

Communication via Social AR: TikTok filters are communicative by design. Using one is a performance. The filter provides the prop; the user must frame, narrate, and express the interaction, applying digital storytelling skills to turn play into cultural advocacy (Zhou & Wang, 2022).

Cognitive Load and Intuitive Design: Effective educational AR requires balancing engagement with learning. The literature suggests that overly complex interactions can overwhelm users, while intuitive, reward-based interactions (like tap-to-animate) sustain attention and reinforce learning goals (Mayer, 2021). The VisionARies filter's design

employs a universal tap gesture across three distinct games, which aligns with principles of reduced cognitive load (Mayer, 2021), allowing users to focus on appreciating each game's unique motion and cultural meaning.

B. The Identified Gap: From Passive Viewing to Active Skill-Based Interaction in Social AR

While many cultural institutions use social media AR filters for branding or photo opportunities, a significant gap exists in filters deliberately designed to apply thinking skills and foster user-generated heritage communication.

Most heritage-related filters are passive. They do not require cognitive engagement with the cultural object's function or rules.

There is a lack of filters that simulate the core mechanic of a traditional game, requiring the user to perform an action that mirrors real-world skill (like timing a takraw kick or spinning a top).

Few projects leverage the social sharing aspect of TikTok as an intentional outcome of this interactive process, where the act of playing the filter becomes the content for communicative cultural advocacy.

C. How VisionARies Fills the Gap: A Skill-Based Interactive Filter Framework

The VisionARies TikTok AR filter directly addresses this gap by transforming three traditional Malaysian games (Wau, Gasing, Sepak Takraw) into an interactive experience that applies user skills and prompts communication.

Application of Thinking & Motor Skills: Each game in the filter applies a different cognitive and tactile engagement:

Sepak Takraw (Tap to Bounce): Users learn timing and rhythm to keep the takraw in the air, applying predictive thinking and motor timing.

Gasing (Tap to Spin & Sustain): Requires repeated tapping to initiate and maintain momentum, teaching concepts of force and persistence.

Wau (Tap to Lift & Fly): Focuses on achieving and controlling ascent, engaging cause-effect reasoning.

This interactive design moves beyond viewing; it requires focused attention, pattern recognition, and kinetic learning to interact "correctly" with each cultural object.

Application of Communication & Creative Skills: The filter creates shareable content. Users are encouraged to narrate attempts, showcase "high scores," or explain cultural context, applying creative digital communication and interpretation skills for peer-to-peer education.

Bridging the Skill Loop: VisionARies creates a direct link: the cognitive skills applied in the filter become the subject of digital communication. It provides an interactive "skill story," demonstrating how social AR can activate thinking and empower users as heritage communicators.

III. METHODOLOGY

This project involved the design and development of an Augmented Reality (AR) guide for playing heritage games. The project was carried out in collaboration with Badan Warisan Malaysia, where the AR guide will be used as an interactive educational tool to enhance visitors' understanding and engagement with traditional games through digital technology.

A. Participants

Participants consisted of cultural heritage stakeholders from Badan Warisan Malaysia and end users involved in usability testing. End users were university students aged 18–25 years recruited through convenience sampling, reflecting the primary TikTok user demographic.

B. Procedure

Both qualitative and quantitative data were collected through user testing sessions and a Google Form questionnaire administered to university students at Universiti Malaya. Qualitative responses were analyzed thematically to identify usability and content-related issues, while quantitative data were analyzed descriptively to assess overall interaction effectiveness.

A user-centered and iterative development approach was used in this project. The project was conducted over a fixed development period of approximately 14 weeks. The project timeline was divided into

several phases: planning and research, design and prototyping, development, testing and refinement, and final deployment. Early weeks focused on requirement gathering and content research, including a site visit to Badan Warisan Malaysia to observe the traditional games and consult with stakeholders to analyse the issues faced by them, followed by asset creation and AR interaction development. The final weeks were dedicated to testing, bug fixing, and documentation.

Development activities were primarily carried out remotely using online collaboration tools, while consultation and feedback sessions were conducted in collaboration with representatives from Badan Warisan Malaysia via email.

C. Materials

The ICT solution developed in this project was an AR-based interactive Tiktok filter created using Effecthouse. The filter includes 3D models, game instructions, and user interactions to simulate traditional games. TikTok was selected as the deployment platform due to its wide user reach, accessibility, and suitability for informal learning environments. The final AR TikTok filter will be implemented on-site at Badan Warisan Malaysia as well as accessed by the public through the TikTok platform.

This project did not involve any direct costs, as Effect House was used as a free development and publishing platform. No paid software, licenses, or additional resources were required.

IV. RESULTS AND EVALUATIONS

This section presents the evaluation of VisionARies, focusing on user demographics, system usability and cultural impact. Using both quantitative and qualitative methods, the evaluation examines user acceptance, visual quality, ease of use and overall engagement. The findings provide insight into the application's effectiveness as an educational AR tool and highlight areas for future improvement and expansion.

A. Participants Demographics

The evaluation was conducted with a sample of 27 participants, primarily aged 18–23. This cohort represents the "digital native" demographic most

likely to engage with augmented reality (AR) on social media platforms like TikTok. All participants (100%) were Malaysian nationals, which was essential for validating the cultural authenticity of the digitalization of traditional games. Data analysis revealed a gender distribution of 66.7% male (n=17) and 25.9% female (n=8). Regarding the technical environment, 77.8% of participants used Android devices, while 22.2% used iOS. This distribution underscores the necessity of performance optimization for various hardware tiers, specifically ensuring stability on mid-to-low-range Android devices.

B. Quantitative Evaluation (Evaluation Matrix)

The quantitative assessment utilized a 5-point Likert scale across five core metrics to measure the application's effectiveness. The VisionARies AR filter achieved an aggregate score of 4.48 out of 5.0, indicating exceptional user acceptance.

The results indicate strong positive feedback across all aspects. Visual appeal scored the highest with an average of 4.7 out of 5, showing that users loved the graphics and overall "wau" look. Interactivity and user experience followed closely at 4.6, confirming that the AR features worked well. Educational value also scored 4.6, indicating clear understanding of the purpose. Clarity and ease of use received a slightly lower score of 4.1 due to minor text size and scaling issues. Overall satisfaction remained high at 4.4, with users willing to reuse the system, especially if gamified.

C. Qualitative Feedback and Proof of Activities

Qualitative feedback provided deeper insight into the user experience, revealing three recurring themes: positive reception, technical pain points, and a future "wishlist." Participants described the filter as "visually compelling" and "smooth," noting that the educational purpose, reducing the need for repetitive manual explanations of traditional games, was immediately apparent.

However, several areas for improvement were identified. Users noted that on-screen text was occasionally too small to read comfortably and that audio levels were perceived as excessively loud by some. Furthermore, the "Wishlist" revealed a

significant demand for gamification. Users expressed high interest in competitive elements such as scoring systems, leaderboards, and "battle modes," as well as content expansion into other traditional games like Congkak.

D. Impact & SDG Achievements

The VisionARies project demonstrates a tangible contribution to the United Nations Sustainable Development Goals (SDGs), bridging the gap between modern technology and cultural heritage.

SDG 4 (Quality Education): The filter functions as an interactive educational tool. By transforming abstract heritage into an immersive simulation, the project facilitates knowledge transfer to Generation Z in a format that suits their digital learning preferences.

SDG 9 (Industry, Innovation, and Infrastructure): The project uses innovative AR infrastructure to address the challenge of cultural erosion. Participants' recognition of the tool as "unique" and "interactive" confirms its status as a novel digital solution.

SDG 11 (Sustainable Cities and Communities): By digitizing the "Wau" and other traditional elements, the project actively safeguards intangible cultural heritage (Target 11.4). Users explicitly recognized it as a "vital first step" toward preserving Malaysian identity in the digital age.

E. Summary of Evaluation

In summary, the UAT results confirm that VisionARies is a technically robust and culturally significant application. The 100% recommendation likelihood among participants suggests a high potential for viral growth on TikTok. Future development will prioritize the integration of gamification features and UI refinements to address the identified pain points in scaling and readability.

V. CONCLUSION

The VisionARies project successfully bridges cultural preservation and modern technology through an interactive TikTok AR filter featuring Wau, Gasing, and Sepak Takraw. The project met its three key objectives which are to reduce staff workload via automated AR guidance, to increase

visitor awareness of traditional games and to enhance engagement through interactive AR. User testing with 20 Malaysian youths confirmed success, with an aggregate satisfaction score of 4.33/5, high educational value (4.6/5), and strong interactivity (4.6/5). Feedback specifically noted the filter's potential to reduce repetitive staff explanations, fulfilling the operational efficiency goal.

By using a simple tap-based interface, the filter transforms passive viewing into active skill-based interaction. Users apply predictive thinking, cause-effect reasoning, and motor timing while engaging with each game's mechanics. This design also encourages users to create and share content, applying digital storytelling and multimodal communication skills in practice. The project demonstrates that technology can be designed not just to deliver content, but to actively develop users' cognitive and communicative abilities.

Recommendations for practice include integrating the filter into heritage site tours and school programs, training volunteers in AR facilitation, and using analytics to track cultural reach. For communication skills training, VisionARies shows that gamified, experiential AR modules can effectively build both digital literacy and soft skills like public explanation and adaptive instruction.

VI. FUTURE WORKS

For future work and sustainability, VisionARies has strong business potential. It can be licensed to tourism boards, expanded into a paid version with a scoring system, and enhanced with more traditional games like Congkak, as well as partnerships with cultural festivals. The next steps focus on user needs by adding a scoring system and leaderboard, including the Congkak game, improving ease of use such as bigger text and better screen scaling, and installing the system at partner locations to track user interest and long-term engagement.

VII. REFLECTIONS

The project underscored the value of interdisciplinary collaboration and user-centered design. Balancing authentic cultural representation with engaging gameplay emerged as a key challenge. Ultimately, VisionARies illustrates how social AR can turn visitors into active heritage

communicators, using technology to sustain intangible culture in the digital age.

VIII. REFERENCE

[1] Ibrahim, N., Ali, N. M., & Mohd Yatim, N. F. (2018). AR in Informal Learning Environments: A Review of Research. *Journal of Educational Technology & Society*, 21(3), 222-231.

[2] Mayer, R. E. (2021). *Multimedia learning* (3rd ed.). Cambridge University Press.

[3] Ruiz, S., Guan, T., & Moghavvemi, S. (2021). Social AR: Transforming how we connect, learn, and play. *IEEE Computer Graphics and Applications*, 41(4), 23-33.

[4] Zhou, Y., & Wang, P. (2022). Heritage communication on short-video platforms: The case of TikTok. *Journal of Cultural Heritage Management and Sustainable Development*, 12(1), 89-104.

IX. APPENDIX A: PROOF OF ACTIVITIES



Visitation to BWM Heritage

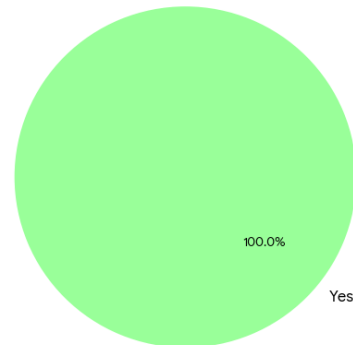


Discussion for planning and creating AR effects

Timestamp	Age	Gender	Nat	What is your role?	How do you think?	What was your first?	How do you think?	Did the filter?	How engage?	How is?	What is the OR?	Are there?
10/01/2024 11	22	Male	Malay	UI	5	5	Very amazing	5	None at all	To understand it	Yes	5
10/01/2024 11	21	Male	Malay	Android	4	4	It's very interesting	4	No	To make it as a filter	Yes	4
10/01/2024 11	21	Male	Malay	Android	4	5	Very interesting and easy	4	Everything	To introduce the filter	Yes	5
10/01/2024 11	20	Male	Malay	Android	2	4	It's a first step for me	4	To make it as a filter	To make it as a filter	Yes	4
10/01/2024 11	20	Female	Malay	IOS	5	5	Very amazing	5	Not at all	None	Yes	5

Sample of responses for data collection

Would Recommend



Response of recommendation from users

Metric	Estimated Avg (out of 5)	Interpretation
Visual Appeal	4.7	Users love the graphics and the "Wau" look.
Interactivity	4.6	The AR element is working well.
Clarity/Ease of Use	4.1	Good, but slight issues with scaling and text size.
Educational Value	4.6	Users clearly understood the purpose (Heritage/Education).
Future Interest	4.4	High willingness to use it again, especially if gamified.

Evaluation matrix result

X. APPENDIX B: INTERVIEW QUESTIONS

1. What is your age?
2. What is your gender?
3. What is your nationality?
4. What device did you use to test the filter? (iOS/Android)?
5. How easy was it to find and open the AR filter? (1–5)
6. Were the instructions clear and easy to understand? (1–5)
7. What was your first impression of the filter?
8. Did the filter respond accurately to your movements (face, hands, body, environment)? (1–5)
9. Were any interactions confusing or unclear?
10. What do you think is the main purpose of this filter?

11. Did the filter meet that purpose?
(Yes/No)

12. How engaging did you find the
filter? (1–5)

13. How likely are you to share a
video using this filter? (1–5)

14. What is the ONE thing you
would improve about this filter?

15. Are there any features you
expected but did not see?